



NODD

Technical Design Report

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PRELIMINARY PIXEL BASELINE DRAFT · 2026-10-06

Typeset with pdf $\LaTeX$ in Palatino (newpx) with TeX Gyre Heros headings and Inconsolata for code.

Source: self-contained TDR review sources in the nODD repository, derived from the pinned Overleaf template. Pixel parameters and evidence come from DES013, DES017–019 and the corresponding frozen validation records. This preliminary draft is not an engineering sign-off.

Emblem: the Open Data Detector logo, wordmark removed.

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Introduction

nODD develops an experiment-independent reference detector with explicit module, support, cooling and service hypotheses. The preliminary pixel baseline combines an unchanged barrel with radially oriented single- and quad-module endcaps. The selected endcap layout removes only complete inner rings proved inaccessible within the stated track and luminous-region envelope.

This manuscript records a preliminary design and an isolated DD4hep implementation. Selection by the project maintainer establishes the comparison baseline; it does not establish manufacturing readiness, engineering sign-off or full-detector acceptance. Detector parameters, derivations and validation evidence remain canonical in the nODD Git repository. The sources and generated assets in this review copy are self-contained for pdfLaTeX and can be transferred to Overleaf by a deliberate manual publication step.

[Chapter 2](#) describes the selected layout, local support and services, implementation and measured construction checks. Other detector systems and performance chapters remain outside this update.

Preliminary pixel detector

2.1 Selected baseline and applicability

The preliminary baseline combines the unchanged four-layer DES013 barrel with nine discs at each end. Endcaps use the DES017 mixed single/quad template, trimmed by the DES018 luminous-region and track-envelope proof. The isolated DD4hep compact implements this selection under DES019; the old quad-only compact remains a historical control. Baseline selection does not grant engineering sign-off [1, 3, 4, 5].

The combined inventory contains 5 362 modules and 11 806 active chip areas. The unchanged barrel contributes 3 050 modules and 6 794 active areas; the two endcaps contribute 2 312 modules and 5 012 active areas [1, 5]. These counts describe sensitive islands. Each quad shares one silicon substrate across four active islands.

The trimming proof applies to primary charged tracks with $p_T \geq 1$ GeV and $|\eta| \leq 4$, luminous z within ± 150 mm, and transverse vertex coordinates $x, y \in [-1, 1]$ mm. It assumes a uniform axial field with magnitude at most 4 T, vacuum propagation, and the first outward half-turn before leaving the comparison host. The closed eta boundary conservatively implements the requested $|\eta| < 4$ [4]. This is a bounded geometric hypothesis, not a measured beam profile or full physical acceptance claim. Lower-momentum, secondary, scattered and returning trajectories require further study.

2.2 Unchanged barrel layout

The barrel retains the selected DES013 packed layout, including module IDs, stave staggering, supports, cooling and services. Inner layers use singles; outer layers use quads. Its active chip dimensions are 19.2 mm tangentially and 20 mm along the beam. Barrel sensor outlines and periphery orientation remain those of the packed reference, rather than adopting the

different endcap outlines [1]. Native regression compares the complete barrel entity/material inventory with its original export [5].

Table 2.1. Unchanged barrel inventory from the frozen DES013 packed layout. Radius denotes the nominal layer datum, not every staggered sensor plane; the last column bounds the actual active-area extent [1].

Layer	r_d [mm]	Family	Staves	Modules	Chips	$ z _{\max}$ [mm]
1	34	single	12	636	636	545.600
2	60	single	22	1166	1166	545.600
3	106	quad	18	468	1872	540.400
4	182	quad	30	780	3120	540.400

2.3 Endcap layout and inner-ring removal

The full mixed template has four single-module rings followed by two quad-module rings: six physical rings containing eight radial chip rows. Modules follow local tangential and radial axes, with alternating support faces and frozen height levels. Signed transverse compensation is retained at every disc datum [3, 5].

For positive datum z_d , luminous half-length L and frozen local height δz , define $a = (z_d^2 - L^2)/z_d$. Module centres obey $(x, y) = (x_0, y_0)(1 + \delta z/a)$ and sensor plane $z_d + \delta z$. Negative placements reflect z and retain the same transverse positions. Filtering never recolours, moves or compacts survivors [3, 4, 5].

Whole inner rings are removed only when every active-area corner lies below the continuous lower reachable-radius bound, including curvature and the vertex box, with a 0.1 mm retention buffer. A reachable witness on the next ring prevents its removal. The selected prefix is maximal within this frozen catalogue, rather than a global layout optimum [4].

The schedule removes 424 single modules/chips across both ends relative to the full mixed template, saving 1 139.712 W nominal heat at the provisional 2.688 W per chip [3, 4]. Plate and carrier envelopes stay full size; missing sensor rings do not imply a plate mass saving.

Table 2.2. Trimmed mixed endcap at each end. Displayed datums are rounded to 0.001 mm; DD4hep retains source precision. Ring numbers refer to the original template. Generated from DES017/018 [3, 4].

Disc	$ z_d $ [mm]	Removed	Single	Quad	Chips	Circuits
1	615.200	none	102	50	302	16
2	794.414	none	102	50	302	16
3	1046.270	none	102	50	302	16
4	1333.096	none	102	50	302	16
5	1645.288	1	84	50	284	14
6	1977.808	1	84	50	284	14
7	2327.479	1–2	62	50	262	12
8	2692.091	1–3	34	50	234	10
9	3070.000	1–3	34	50	234	10

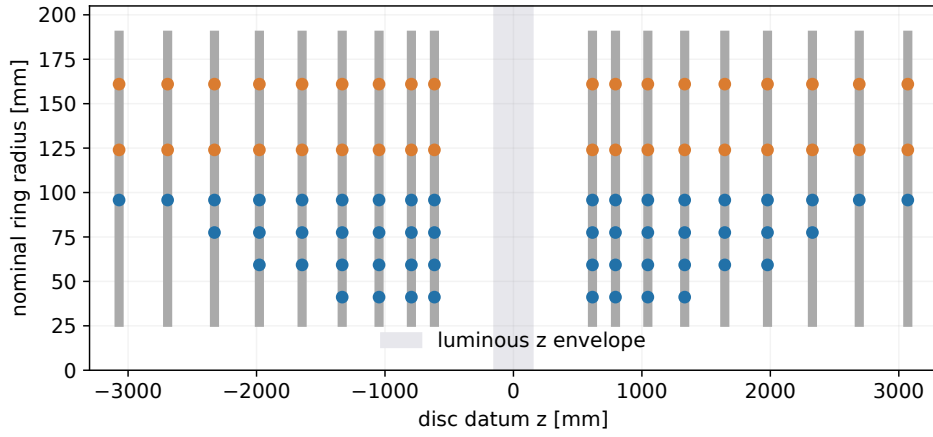


Figure 2.1. Disc datums and nominal ring radii. Singles are blue and quads orange; grey lines show the unchanged plate annulus. The shaded band is the luminous z envelope. Points indicate nominal radii, not individual compensated planes. The barrel is omitted to keep the trimming schedule visible.

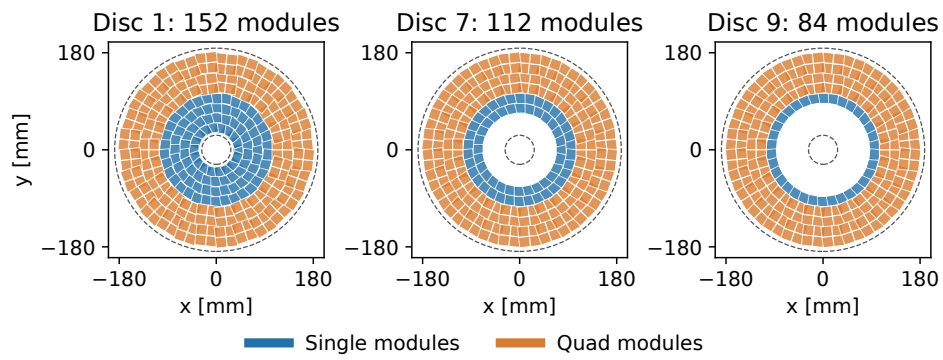


Figure 2.2. Active-area projections for positive discs 1, 7 and 9. Blue denotes single modules and orange quads; dashed circles mark plate radii. Projection overlaps are not volume collisions because modules use alternating faces and height levels. Inherited coverage gaps remain; hermeticity is not asserted.

2.4 Modules and readout orientation

Each endcap active island is 20 mm tangentially by 19.2 mm radially, with conditional 0.1 mm guard and 0.2 mm inactive seams. Shared silicon outlines are 20.2×19.4 for singles and $40.4 \times 38.8 \text{ mm}^2$ for quads. Occupied body reservations are 20.4×21.4 and $43.2 \times 44.2 \text{ mm}^2$, respectively, with 1 mm thickness [2, 5]. These are project hypotheses, not procurement specifications.

The reused stack has 0.15 mm sensor and ASIC silicon, 0.02 mm bump stand-off and inherited flex/glue layers. Nominal $50 \mu\text{m}$ square segmentation remains unchanged and provisional [1]. The physical tangential/radial convention supports future non-square segmentation without rotating measurement interpretation with global azimuth. Frame signs establish a proper rotation and place the ASIC side toward its pickup; tangential/radial dimensions are not swapped [5].

2.5 Local support, cooling and mounting

The common support retains $r = 27\text{--}188.5 \text{ mm}$ and the 6.3 mm sandwich: two 0.15 mm CFRP skins around a 6 mm foam core. Three inherited tongues couple to rails and the carrier, whose outer radius remains 232 mm. Coupling details and installation tolerances are unresolved [3, 5].

Every retained chip has a 0.45 mm cradle/graphite/insulator/TIM/bond pickup. Graphite stems have $6 \times 8 \text{ mm}^2$ cross-section; quad stems move 1 mm toward the module centre. Raised stems follow frozen 1.65 mm level spacing. Removed rows delete their pickups and cooling demand. Closing unused skin/core windows while preserving structural and thermal interfaces remains a manufacturing task [3, 4, 5].

Each retained chip row uses two half-ring circuits, giving 248 circuits across both ends. Tube outer diameter is 2.8 mm, with 0.15 mm titanium wall; core routing planes lie at $\pm 1.5 \text{ mm}$. Transport feed/return diameter stays 4 mm [3, 5]. Flow estimates inherit nominal/stress power and boiling-quality hypotheses, not measured hydraulics.

DD4hep uses complete titanium/ CO_2 tori to preserve summed half-ring length. Radial legs and thermal inserts use an explicit effective core allocation; torus daughters displace the core. Skin/cradle window inventories replace CFRP with graphite once per chip. Bend allowances, clips, mount hardware and retained flex inventories are normalized into local service cells. Constituent volumes and masses are conserved without stacking full foam and occupied solids [5]. This allocation differs from DES017's approximate mass proxy; it is not detailed CAD, hydraulic connectivity or a manufactured bill of materials.

2.6 Services and qualification limits

Collectors retain $r = 27\text{--}190$ mm; the trunk remains $r = 192\text{--}231.7$ mm and flange neck ends at 222 mm. Heterogeneous inventories use each disc's actual chains, links and chip-row circuits. The first eight discs join the trunk; the last follows the inherited bypass [4, 5].

Reference packing utilization is 1.031 7 in the positive trunk and 1.397 2 at its neck. Negative values are 1.028 2 and 1.392 6. All exceed unity; conservative/stress scenarios also fail [4, 5]. Effective cells preserve material demand but do not establish a feasible packed route. Connectors, manifolds, flange passage, bypass and cable topology require further engineering. Service radii are not enlarged to conceal these failures.

2.7 Implementation evidence and next steps

Native DD4hep construction checked all 11 806 active placements, signed axes, normals, dimensions, unique identifiers and readouts. Maximum centre residual is 4.559×10^{-13} mm. Exclusive material mass/volume and sampled radiation/interaction-length scans pass. ROOT finds no overlaps at 1×10^{-5} mm; ROOT and portable nodehammer roundtrips pass. Geant4 conversion and FTFP_BERT initialization pass with seed42 and zero transported events [5].

Simulation material inventory totals 202.751 kg, including effective shared services; this is not a manufactured mass estimate. Native execution used the recorded dirty implementation tree, identified by actual compact/input/producer/plugin hashes. Its execution revision is distinct from the enclosing deliverable commit [5].

DD4hep encodes template ID offsets without compacting survivors and retains their original global source IDs in the inventory. The inherited DES018 study compared 1 906 analytic and 182 in-scope native tracks per variant, without removing an in-scope baseline hit [4]. This is not a DD4hep-to-ACTS conversion test. Material transport, realized field map, digitisation, reconstruction, alignment and efficiency/resolution studies remain.

Original coverage gaps persist. The 0.5 mm guard and warm -35°C thermal controls fail [3, 4]. Thermal contacts, two-phase stability, mechanical tolerances, window closure and routing are principal engineering gates. The model provides a reproducible preliminary comparison baseline while these questions remain open.

Bibliography

- [1] nODD, DES013 barrel and inherited DES012 module stack. [Frozen configuration and evidence pins](#).
- [2] nODD, DES016 radial module hypotheses. [Islands, guard, seams and occupied-body dimensions](#).
- [3] nODD, DES017 mixed disc and support comparison. [Results and linked inputs, thermal controls and native evidence](#).
- [4] nODD, DES018 eta-limited disc apertures. [Exclusion proof, inventories and analytic/ ACTS validation](#).
- [5] nODD, DES019 trimmed mixed DD4hep baseline. [Contract and classifications; Native results, service ledger and execution hashes](#).